

Computer Networks Project

Report

TOPIC : NETWORK TOPOLGY

1. Introduction

We made a **departmental network setup** for an **organization**. It includes **three departments**: **Computer Science**, **HR**, and **Management**.
Each department has its own **switch (S1, S2, S3)** and connected **PCs**.

- **PC-A** and **PC-N** belong to the **Computer Science Department** via **Switch S1**.
- **PC-D** is part of the **HR Department** and connects through **Switch S2**.
- **PC-C** and a **Web Server** are in the **Management Department** via **Switch S3**.

The **switches** are linked to **routers (R1 and R2)** using **Gigabit Ethernet**.
Fast Ethernet interfaces connect **PCs and servers** to switches.

VLANs (Virtual Local Area Networks) are used to **separate departmental traffic** and provide **security**.

This network design supports:

- **Routing**
- **NAT (Network Address Translation)**
- **ACLs (Access Control Lists)**
- **Secure communication**

2. Objectives

- To understand the role of switches in a network.
- To learn basic switch configuration commands.
- To implement VLANs for network segmentation.
- To secure switch access using passwords.
- To verify connectivity between devices connected to the switch.

3. Tools and Equipment

- Cisco Packet Tracer / GNS3 (simulation software) or physical Cisco switches.
- PCs or laptops for configuration.
- Ethernet cables for physical connectivity.

4. Switch Configuration Steps

1. **Accessing the Switch**
    a. Connect via console cable or simulation software.
    b. Enter privileged EXEC mode using `enable`.
2. **Basic Configuration**
    a. Set hostname:

Code
```
Switch> enable
Switch# configure terminal
Switch(config)# hostname Lab Switch
```

    b. Configure passwords:

Code
```
Switch(config)# enable secret my password
Switch(config)# line console 0
Switch(config-line)# password console pass
Switch(config-line)# login
```

3. **VLAN Configuration**
    a. Create VLANs:

Code
```
Switch(config)# vlan 10
Switch(config-vlan)# name Sales
Switch(config)# vlan 20
Switch(config-vlan)# name HR
```

    b. Assign VLANs to interfaces:

Code
```
Switch(config)# interface fastEthernet 0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
```

4. **Saving Configuration**

Code

```
Switch# copy running-config startup-config
```

5. Testing and Verification

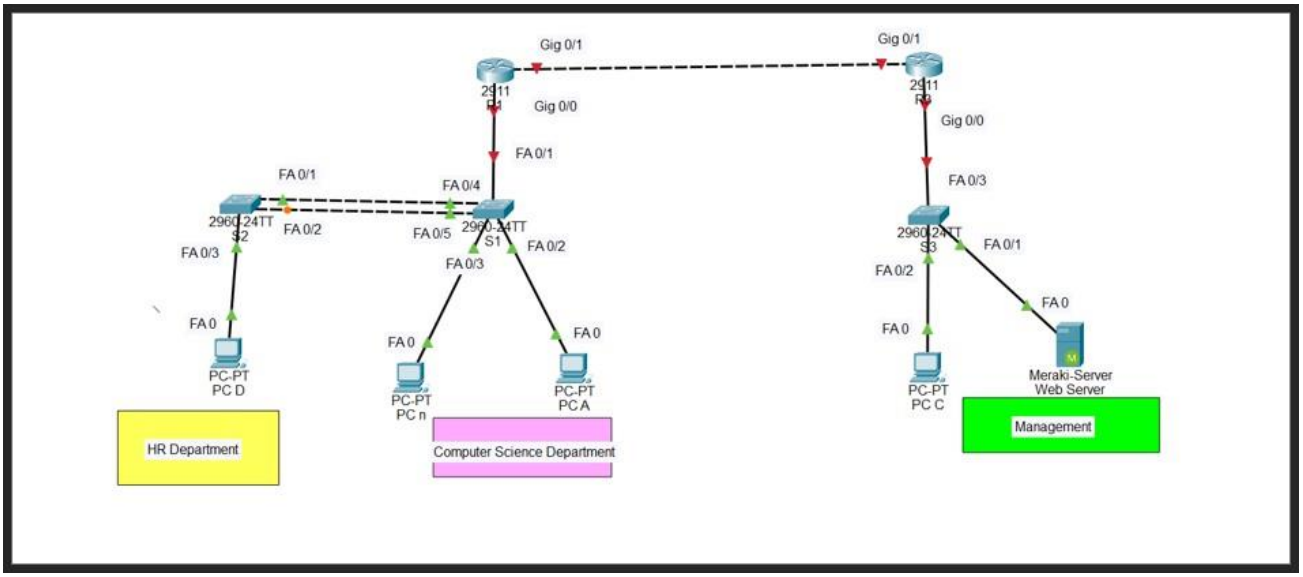
- Use `ping` between PCs in the same VLAN to verify connectivity.
- Use `show vlan brief` to confirm VLAN assignments.
- Use `show running-config` to review applied settings.

6. Results

- Successfully configured switch hostname and passwords.
- VLANs created and assigned to interfaces.
- Connectivity verified between devices in the same VLAN.
- Network segmentation achieved, improving security and efficiency.

DESIGHN :

Here , is a design of our Network Topology



ROUTERS Configuration: (Router 1)

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#no ip domain-lookup
Router(config)#hostname R1
R1(config)#enable secret Ciscopass1
R1(config)#line console 0
R1(config-line)#password Ciscopass11
R1(config-line)#login
R1(config-line)#exit
R1(config)#username admin secret Cisco123
R1(config)#interface gigabitEthernet0/1
R1(config-if)#ip address 172.16.0.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
exit
R1(config)#interface gigabitEthernet0/0
R1(config-if)#ip address 172.16.10.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
R1(config)#end
R1#
```

ROUTER 3

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#no ip domain-lookup
Router(config)#hostname R3
R3(config)#enable secret Ciscopass1
R3(config)#line console 0
R3(config-line)#password Ciscopass11
R3(config-line)#login
R3(config-line)#exit
R3(config)#username admin secret Cisco123
R3(config)#interface gigabitEthernet0/1
R3(config-if)#ip address 172.16.0.2 255.255.255.252
R3(config-if)#no shutdown

R3(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

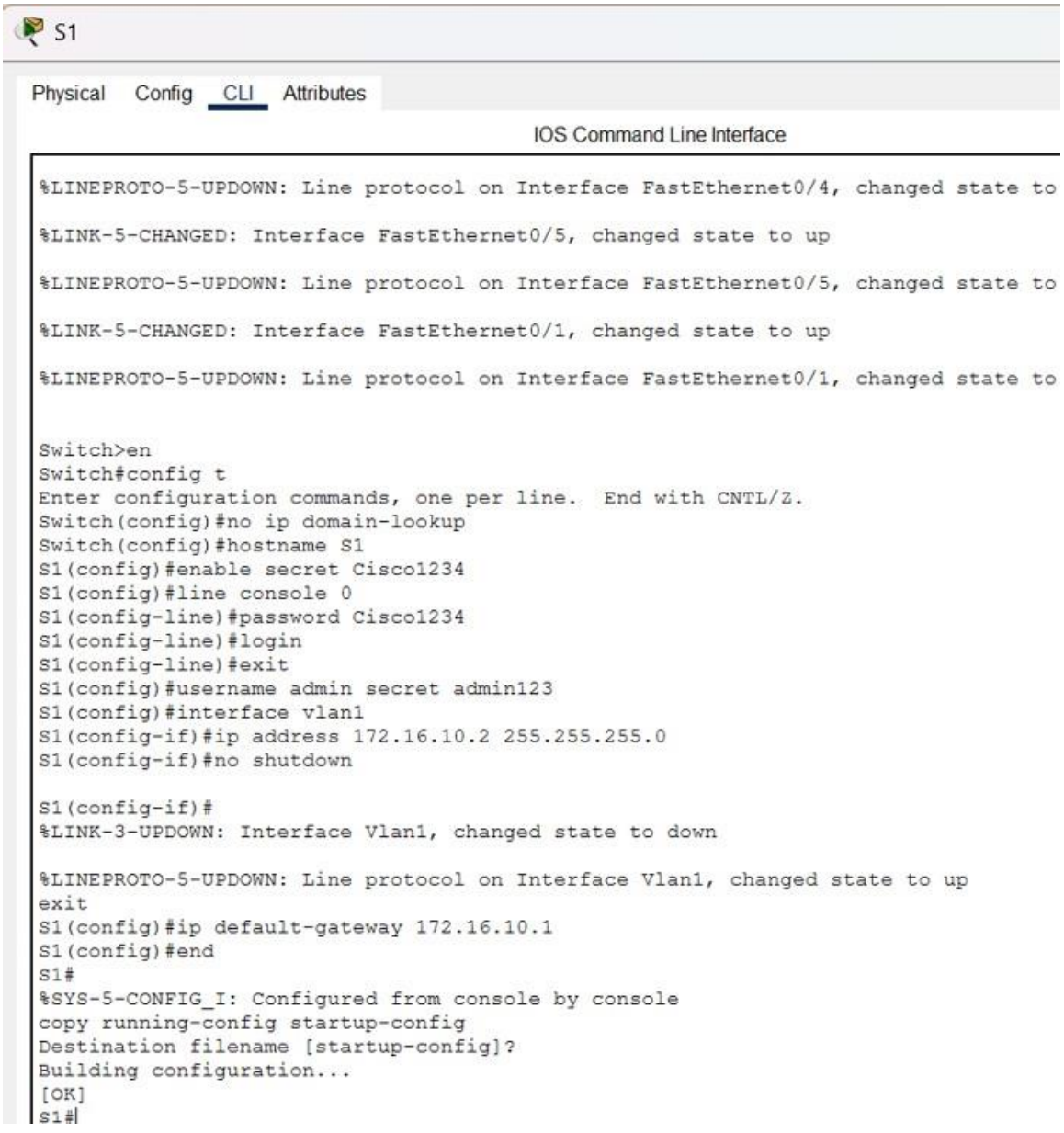
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
R3(config)#interface gigabitEthernet 0/0
R3(config-if)#ip address 172.16.20.1 255.255.255.0
R3(config-if)#no shutdown

R3(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R3(config-if)#exit
R3(config)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console
copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
```

Switches Configuration (Switch 1)




S1
Physical
Config
CLI
Attributes
IOS Command Line Interface
User Access Verification
Password:
S1>configure terminal
^
% Invalid input detected at '^' marker.
S1>config t
^
% Invalid input detected at '^' marker.
S1>en
Password:
S1#en
S1#config t
Enter configuration commands, one per line. End with CNTL/Z.
S1(config)#vtp domain AU-NET
Changing VTP domain name from NULL to AU-NET
S1(config)#%SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain name changed to AU-NET.
S1(config)#vtp mode server
Device mode already VTP SERVER.
S1(config)#vtp password vtp123
Setting device VLAN database password to vtp123
S1(config)#vlan 2
S1(config-vlan)#name Computer_Science
S1(config-vlan)#exit
S1(config)#vlan 3
S1(config-vlan)#name HR
S1(config-vlan)#exit
S1(config)#vlan 4
S1(config-vlan)#name Management
S1(config-vlan)#exit
S1(config)#vlan 5
S1(config-vlan)#name NULL
S1(config-vlan)#exit
S1(config)#

Switch 2


S2
Physical
Config
CLI
Attributes
IOS Command Line Interface
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/
Switch>en
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#no ip domain-lookup
Switch(config)#hostname S2
S2(config)#enable secret Cisco1234
S2(config)#line console 0
S2(config-line)#password Cisco1234
S2(config-line)#login
S2(config-line)#exit
S2(config)#interface vlan 1
S2(config-if)#ip address 172.16.10.3 255.255.255.0
S2(config-if)#no shutdown
S2(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed
exit
S2(config)#ip default-gateway 172.16.10.1
S2(config)#end
S2#
%SYS-5-CONFIG_I: Configured from console by console
copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
S2#

```
Password:
Password:

S2>en
Password:
S2#config t
Enter configuration commands, one per line.  End with CNTL/Z.
S2(config)#vtp domain AU-NET
Domain name already set to AU-NET.
S2(config)#vtp mode client
Setting device to VTP CLIENT mode.
S2(config)#vtp password vtp123
Setting device VLAN database password to vtp123
S2(config)#vlan 2
VTP VLAN configuration not allowed when device is in CLIENT mode.
S2(config)#interface range fa0/1-2
S2(config-if-range)#switchport mode trunk

S2(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

S2(config-if-range)#switchport trunk native vlan 1
S2(config-if-range)#channel-group 1 mode active
%EC-5-L3DONTBNDL2: Fa0/1 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Fa0/2 suspended: LACP currently not enabled on the remote port.
S2(config-if-range)#
Creating a port-channel interface Port-channel 1

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

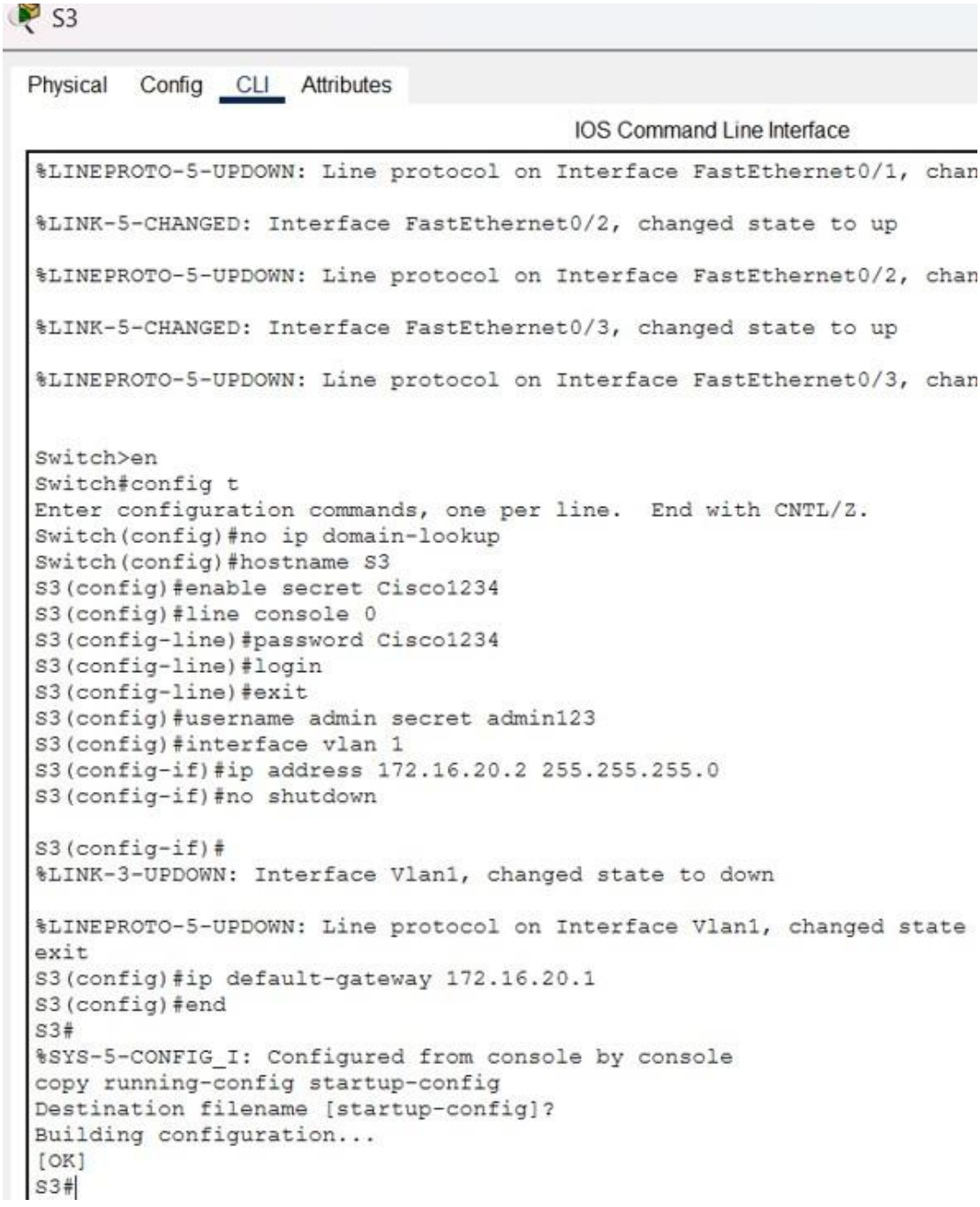
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

S2(config-if-range)#exit
S2(config)#
```

Switch 3


S3
Physical
Config
CLI
Attributes
IOS Command Line Interface
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, chan
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, chan
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, chan
Switch>en
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#no ip domain-lookup
Switch(config)#hostname S3
S3(config)#enable secret Cisco1234
S3(config)#line console 0
S3(config-line)#password Cisco1234
S3(config-line)#login
S3(config-line)#exit
S3(config)#username admin secret admin123
S3(config)#interface vlan 1
S3(config-if)#ip address 172.16.20.2 255.255.255.0
S3(config-if)#no shutdown
S3(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state
exit
S3(config)#ip default-gateway 172.16.20.1
S3(config)#end
S3#
%SYS-5-CONFIG_I: Configured from console by console
copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
S3#

```
User Access Verification

Password:

S3>en
Password:
S3#config t
Enter configuration commands, one per line.  End with CNTL/Z.
S3(config)#vtp domain AU-NET
Changing VTP domain name from NULL to AU-NET
S3(config)#%SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain name changed to AU-NET.
vtp mode client
Setting device to VTP CLIENT mode.
S3(config)#vtp password vtp123
Setting device VLAN database password to vtp123
S3(config)#interface fa0/3
S3(config-if)#switchport mode trunk

S3(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

S3(config-if)#switchport trunk native vlan 1
S3(config-if)#exit
S3(config)#interface fa0/1
S3(config-if)#switchport mode access
S3(config-if)#switchport access vlan 4
S3(config-if)#description Management_Server
S3(config-if)#exit
S3(config)#interface range fa0/4-24
S3(config-if-range)#switchport mode access
S3(config-if-range)#switchport access vlan 5
S3(config-if-range)#description UNUSED_PORT
S3(config-if-range)#shutdown

%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to administratively down
```

```
R1(config-router)#network 172.16.0.0 0.0.0.3 area 0
R1(config-router)#network 172.16.10.0 0.0.0.255 area 0
R1(config-router)#exit
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
copy running-config startup-config
Destination filename [startup-config]?
```

7. Conclusion

This lab Report demonstrated the importance of switch configuration in computer networks. By setting up VLANs and securing access, we ensured proper segmentation and security. The exercise provided hands-on experience with essential commands and reinforced understanding of Layer 2 networking concepts.

…. THE END ….